Preliminary Analysis of Weibo Emoji Usage in Spam Posts of Social Media

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Abstract: In this paper, we introduce our preliminary analysis of emoji being used by spammers in Chinese social media Weibo for the future work on spam detection utilizing pictograms. We assume that analyzing the characteristics of spam comments and the sentiment polarities of emojis used in Weibo comments, could enhance the accuracy of social media spam detection. We summarize characteristics of spam posts and emoji used in these post. From our early findings it can be said that combining sentiment polarities of emojis and neural networks can be expected to achieve better performance for spam detection in the future.

1 Introduction

Today, many people share information about their daily lives with their friends by posting status updates on Facebook, sharing their holiday photos on Instagram or tweeting their views via Twitter or its Chinese counterpart Weibo - the biggest social media network in China that was launched in 2009. Social media data contain a vast amount of valuable information [Li 18a], many studies utilize Weibo data to analyze hot topics and sentiment evolution patterns to gain insights into users' behavior. However, various spam messages often obfuscate the current data-related tasks standing on their way to achieving better accuracy. Furthermore, some users are being mislead by malicious links or message causing financial or reputation losses. To mitigate the damage caused by spam, various classification algorithms have been applied to distinguish spam from non-spams [Dhaka and Mehrota 19, Tang 19]. On the other hand, majority of works on this domain treat emojis as noise, and focus on detecting spammers rather than spam messages. Most papers focus on English spam detection on social media networks such as Twitter. Moreover, we noticed that a lot of spam in Weibo contains many emojis, which are abundant (more specific analysis of this phenomena is discussed in Section 3). Therefore, in this paper, we focus on emojis used in spam content instead of detecting spamming behaviors, and investigate the characteristics of unsolicited or undesirable posts on Weibo. Our working hypothesis is that by analyzing the characteristics of spam comments and the sentiment polarities of emojis used in Weibo posts, it will help enhance the accuracy of social media spam detection in the future.

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2 Related Research

[Liu 17] proposed a fuzzy-based oversampling method that generates synthetic data samples from limited observed samples based on fuzzy-based information decomposition to address the problem of unequal distribution between spam and non-spam classes in Twitter. [Lin 13] designed an automatic online spammer identification system. They claim that their system can effectively detect the spamming behaviors and identify spammers in Weibo. In this paper, instead of focusing on spammers, we concentrate on spam text and emoji types used in the spam comments.

In our previous research, [Li 18b] we analyzed the usage of the emojis with facial expression used on Weibo. We asked 12 Chinese native speakers to label Weibo emojis by applying one of three following categories: positive, negative and humorous. We have confirmed that 23 emojis can be considered more as

Table 1: Examples of Emojis Conveying Humor Typical for Chinese Culture.

Emoji	Humorous	Negative	Positive
	91.7%	8.3%	0.0%
	83.3%	0.0%	16.7%
	58.3%	25.0%	16.7%
	75.0%	8.3%	16.7%
	75.0%	0.0%	25.0%

humorous than positive or negative, some examples shown in Table 1. On this basis, we plan to apply the sentiment polarities of Weibo emoji to analyze the emoji usage of spam comments.

3 Analysis of Weibo Comments Spam

To verify our hypothesis, we collected first 500 Weibo comments containing emojis from the "Trending Topics" category. All comments were posted in February, 2020. Then we asked three Chinese native speakers to annotate them in two categories: "spam" or "not spam". After one annotator labeled polarities of all posts, two other native speakers confirmed the correctness of his annotations. Whenever there was a disagreement, all evaluators decided the final polarity through discussion. The results show that the data set was divided into 367 spam comments and 133 general, non-spam comments. The analysis of Weibo emoji usages and the characteristics of Weibo spam comments, are summarized bellow:

- Some spammers are real users, not bots. Often spammers compromised accounts of existing users and send spam messages as these users. We should focus on spam messages rather than spamming accounts in the future, since a spam content can be sent from an actual user's account.

- Some spam comments are relevant to the topic of a post, which highly increases the difficulty for detection.

  An example shown in Figure 1, in this porn-related advertisement, a spammer commented on Weibo entry about "Happy Star" (Chinese science fiction TV series for children) and wrote a related dirty story on this theme to attract people's attention. To address this problem, a neural network approach considering emojis and context (for example, emoji-aware BERT model) should be considered in the next stage.



Figure 1: A example of spam comment which relevant to the topic.

- Majority of the spam comments describe erotic content and advertising, due to the strict censorship in the Chinese online communities (it is illegal to make sex-related material open to public).

- Another interesting phenomenon we observed is that a great number of spam comments are semantically ill-structured. Spammers replace the sensitive words with other words/symbols (with similar meaning or pronunciation) or separate sensitive words with emojis to avoid being detected (an example shown in Figure 2). It can be problematic to detect such examples of deliberate content concealing.



Translation: "username: charming/tender [awesome] school belle [Clapping Hands] [awesome] @username"

Figure 2: A example of erotic message which avoids automatic detection by inserting a pictogram inside the text.

Table 2: Occurrence of emojis used in 500 randomly chosen Weibo comments on trending topics.

Emoji	Spam	Non-spam
❤️	863	19
🌹	247	2
转发	150	0
👈	87	0
😞	60	1
🤦	6	41
🐶	0	83

- Since many spam posts contain emojis, it is necessary to analyze how the Weibo emojis are used in spam comments. We selected several representative emojis and listed how many times they were used in our data set. The results (see Table 2) show that spam messages rarely use humorous emoji which were annotated in our previous research. This phenomenon suggests a probable use for applying humorous sentiment polarities of emojis to automatically classify spam and non-spam comments in the future. Furthermore, we noticed that spammers tend to use emojis without facial expression, such as ❤️ and 🌹, we also observed that pointing fingers (👈) are almost always used to point to links or accounts inviting users to click.

4 Conclusions and Future Work

In this paper, we introduced our preliminary analysis on usages of emojis in the context of spam, and investigated the characteristics of spam comments on Chinese social media network, Weibo. We introduced our hypothesis that some emojis are characteristic and uncharacteristic for spam content. To verify our hypothesis, we collected 500 Weibo comments data, analyzed Weibo emoji usage, and listed characteristics of Weibo spam comments. We believe that by applying sentiment polarities of emojis and neural network, a high-quality spam detection method can be constructed in the future. This preliminary investigation is the first step toward this goal.

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